

NUST 1980 — QC Issues Log (PDF vs CSV Values Verification)

Last updated: 2026-04-30

QC method: `scripts/qc_pdf_vs_csv.py --mode values` — cell-by-cell PDF vs CSV comparison via Claude API

Status: COMPLETE — all 144 Test×Location combos verified

Total cost: ~\$15 (prompt caching active; first call cache-write, subsequent calls cache-read)

Raw output: `output_1980/qc/qc_1980_values.csv`, `output_1980/qc/qc_1980_values_progress.json`

Master Issues Summary

#	Category	Scope	Tests	Action	Status
1	Column swap — Height Fargo/Morden	9 strains × 2 cities	UT-00	Swap + patch from PDF	Confirmed from XLSX
2	Missing data — SeedQuality Morden NULL	9 strains	UT-00	No action	Confirmed genuine
3	Cell errors — SeedQuality (2 cells)	2 cells	UT-00	Patch from PDF	Confirmed
4	Cell error — SeedSize Fargo (1 cell)	1 cell	UT-00	Patch from PDF	Confirmed
5	OCR strain names (5 strains)	Multiple locations	PT-I/II/III	Fix script	DONE
6	Column shift — PT-II Yield (Lafayette, Urbana)	5 cells	PT-II	XLSX verify → patch	Suspected
7	Cell errors — PT-III (2 cells)	2 cells	PT-III	Patch from PDF	Confirmed
8	Missing values — NULLs linked to ghost strains	~11 cells	PT-I/III	Resolves after ISSUE-5	Resolved
9	Cell errors — PT-IV Lexington Oil (2 cells)	2 cells	PT-IV	Manual PDF verify first	Needs verify
10	Missing Maturity — PT-IV Portageville Loam	36 rows	PT-IV	Insert from XLSX tp7 block	Confirmed
11	OCR strain name — M72- 107 space	All 8 locations	UT-0	Merge rows	DONE

#	Category	Scope	Tests	Action	Status
12	OCR strain name — Hardin / Hardin (I) split	3 locations	UT-I	Merge rows	DONE
13	OCR strain name — Pella (TII) → Pella (III)	All 23 locations	UT-II	Merge rows	DONE
14	OCR strain name — U2020325 → U20325	All 23 locations	UT-II	Merge rows	DONE
15	OCR strain name — U566355 → U56355	5 locations	UT-II	Merge rows	DONE
16	OCR strain name — BSR 320 → BSR 302	6 locations	UT-III	Merge rows	DONE
17	OCR strain name — K1035 → K1033	All 21 locations	UT-IV	Merge rows	DONE
18	OCR strain name — Ky75-146-74 variants (3 forms)	15+ locations	UT-IV	Merge rows	DONE
19	OCR strain name — H76-3840 → HC76-3840	5 locations	UT-IV	Merge rows	DONE
20	Column shift — UT-I Lafayette (Lodging/Height/SeedQuality)	11 cells	UT-I	XLSX verify → patch	Suspected
21	Column shift — UT-III S. Charleston Lodging (all strains)	~22 cells	UT-III	XLSX verify → patch	Suspected
22	Column shift — UT-IV Manhattan Lodging (all strains)	~13 cells	UT-IV	XLSX verify → patch	Suspected
23	Column shift — UT-IV Powhattan Yield (4 strains)	4 cells	UT-IV	XLSX verify → patch	Suspected
24	Column shift — UT-II Harrow Yield (3 strains)	3 cells	UT-II	XLSX verify → patch	Suspected
25	Scattered cell errors (17 cells across 6 tests)	17 cells	UT-0/I/II/III/IV	XLSX verify → patch	Needs verify

Action Plan (fix order)

1. **Fix all OCR strain name duplicates** (ISSUES 5, 11–19) — one consolidated fix script
2. **Cross-check XLSX for all suspected column shifts** (ISSUES 1, 6, 20–24) before patching
3. **Manual PDF verify** ISSUE-9 (PT-IV Lexington Oil) and ISSUE-25 scattered cells
4. **Insert missing Maturity rows** for PT-IV Portageville Loam (ISSUE-10)
5. **Apply all confirmed cell patches** in one pass

CATEGORY A — OCR Strain Name Duplicates

All of these follow the same pattern: OCR misread a strain name, creating a ghost row. Traits end up split between the canonical name and the ghost name. Fix by merging ghost row into canonical and deleting ghost.

ISSUE-5 · Five OCR strain errors — PT-I, PT-II, PT-III

Ghost name in CSV	Correct name	Tests/Locations	Traits affected
A73D29	A75D29	PT-II — all 9 locations	YieldBuA missing for A75D29
Cnome	Gnome	PT-II/Arlington_WI	YieldBuA missing for Gnome
H277-878	HC77-878	PT-II — Hoytville, Lafayette, Urbana	Protein, Oil
A790134034	A79-134034	PT-I — Corwith, Lamberton	SeedSize
L77-709	L78-709	PT-III — 7 locations	Yield, Lodging, Height, SeedSize, Protein, Oil

Fix: `fixes/fix_strain_ocr_1980.py` — already written; extend with new entries below.

ISSUE-11 · M72- 107 (extra space) → M72-107 — UT-0, all 8 locations

Traits split across two rows. Merge under M72-107.

ISSUE-12 · Hardin / Hardin (I) split — UT-I, Arlington, Dundee, Ithaca

Two rows for the same strain. Hardin (I) is the correct UT-I designation. Merge, consolidating Maturity and YieldBuA nulls.

ISSUE-13 · Pella (TII) → Pella (III) — UT-II, all 23 locations

(III) OCR'd as (TII). YieldBuA under ghost row, YieldRank/Maturity under canonical. Also affects SeedQuality and SeedSize at some locations. Merge into Pella (III).

ISSUE-14 · U2020325 → U20325 — UT-II, all 23 locations

Doubled 20 in strain number. YieldBuA under ghost row, rank/maturity under canonical. Merge into U20325.

ISSUE-15 · U566355 → U56355 — UT-II, 5 locations (Adelphia, Hoytville, Landisville, Ridgetown, Wooster)

Extra 6. YieldRank under ghost row. Merge into U56355.

ISSUE-16 · BSR 320 → BSR 302 — UT-III, 6 locations (Adelphia, Clarksville, Hoytville, Landisville, Lexington, S. Charleston)

Digit transposition (302 → 320). Maturity and YieldBuA under ghost row. Merge into BSR 302.

ISSUE-17 · K1035 → K1033 — UT-IV, all 21 locations

K1033 misread as K1035. Height and SeedQuality appear under ghost row; Yield/Rank/Maturity under canonical. Merge Height/SeedQuality into K1033, remove K1035.

ISSUE-18 · Ky75-146-74 variants — UT-IV, 15+ locations

Three ghost forms created by OCR:

- KG75-146-74 — most locations
- K675-146-74 — Lubbock, Powhattan, Portageville, Novelty
- K775-146-74 — Portageville Clay, Portageville Loam, Manhattan

YieldBuA split from YieldRank/Maturity across rows. Merge all variants into Ky75-146-74.

ISSUE-19 · H76-3840 → HC76-3840 — UT-IV, 5 locations (Eldorado, Lafayette, Manhattan, Portageville Loam, Queenstown)

HC OCR'd as H. Protein/Oil appear under ghost row. Merge into HC76-3840.

CATEGORY B — Systematic Column/Location Shifts

These require XLSX raw cell cross-check before patching. Pattern is consistent with data entry errors in the source XLSX (values entered under wrong column header).

ISSUE-1 · Height Fargo/Morden swap — UT-00, all 9 strains (*CONFIRMED from XLSX*)

Root cause: tp10 Height block has column order Morden → Brandon → Fargo (swapped vs tp6–tp9 which use Fargo → Morden → Brandon). Original data values were entered in the old Fargo-first order under the swapped headers.

Evidence: XLSX "Fargo" column = PDF Morden for all 9 strains (perfect match). XLSX "Morden" column ≠ PDF Fargo.

Strain	XLSX "Morden"	XLSX "Fargo"	PDF Morden	PDF Fargo
Clay (0)	20	26	26 ✓	26
Maple Arrow	22	28	28 ✓	28
Maple Presto	21	22	22 ✓	20
McCall	23	27	27 ✓	27

Strain	XLSX "Morden"	XLSX "Fargo"	PDF Morden	PDF Fargo
Portage (00)	21	24	24 ✓	27
M71-148	24	26	26 ✓	26
OT80-1	23	24	24 ✓	26
OT80-2	22	23	23 ✓	23
OT80-3	22	24	24 ✓	25

Fix: Swap Fargo↔Morden Height values; use PDF ground truth for Fargo. Write [fixes/fix_height_swap_1980.py](#).

ISSUE-20 · UT-I / Lafayette_IN — Lodging, Height, SeedQuality, SeedSize (multiple strains)

Multiple traits wrong at a single location — consistent with column mis-assignment in the XLSX.

Strain	Trait	CSV	PDF
Hardin	Lodging	1.5	1.2
Hodgson 78 (I)	Lodging	1.5	1.9
Evans (0)	Height	24	36
Hardin	Height	35	37
M72-3	Height	26	29
Corsoy 79 (II)	SeedQuality	1.5	2.0
Evans (0)	SeedQuality	2.5	3.0
Hodgson 78 (I)	SeedQuality	1.5	2.0
M71-80	SeedQuality	1.5	1.8
M75-2	SeedQuality	1.5	1.6
Hardin	SeedSize	14	14.9

Action: Cross-check XLSX tp9/tp10/tp11 blocks for UT-I Lafayette before patching.

ISSUE-21 · UT-III / S. Charleston_OH — Lodging, all strains

All 22+ strains show wrong Lodging values. Likely a single column shift in the XLSX Lodging block for this location. Other traits may also be affected.

Strain	CSV Lodging	PDF Lodging
BSR 302	3.5	3.2
Century (II)	2.3	1.8

Strain	CSV Lodging	PDF Lodging
Cumberland (III)	2.3	2.0
Williams 79	2.0	2.5
Union (IV)	2.7	2.9
(all other strains)	various	varies

Action: Cross-check XLSX Lodging block for UT-III S. Charleston.

ISSUE-22 · UT-IV / Manhattan_KS — Lodging, all strains

Every strain at Manhattan has Lodging ~2–3 in CSV; PDF shows 1.0 for all. One-value-for-all pattern strongly suggests a single wrong column was read for this location.

Action: Cross-check XLSX Lodging block for UT-IV Manhattan.

ISSUE-23 · UT-IV / Powhattan_KS — Yield, 4 strains

Large gaps suggest row or column shift in XLSX yield block:

Strain	CSV Yield	PDF Yield
Williams79 (III)	16.4	13.5
K1033	18.8	16.4
K1041	18.5	13.6
HC76-3840	11.1	16.6

Action: Cross-check XLSX Yield block for UT-IV Powhattan.

ISSUE-24 · UT-II / Harrow_OH — Yield, 3 strains

Two adjacent strain values appear swapped between rows:

Strain	CSV Yield	PDF Yield
A77-211021	60.4	55.4
A78-122031	54.5	60.4
Pella (III)	55.9	51.6

Action: Cross-check XLSX Yield block for UT-II Harrow.

ISSUE-6 · PT-II / Lafayette_IN and Urbana_IL — Yield, multiple strains

Large gaps for Gnome suggest a column shift. Urbana errors are smaller and may be individual cell mistakes.

Test	City	Strain	CSV	PDF
PT-II	Lafayette	Gnome	18.1	51.1
PT-II	Urbana	Century	42.1	43.8
PT-II	Urbana	Gnome	35.6	49.5
PT-II	Urbana	Hardin (I)	36.6	41.9
PT-II	Urbana	Pella (III)	46.9	45.3

Action: Cross-check XLSX Yield block for PT-II Lafayette and Urbana.

CATEGORY C — Individual Cell Errors (PDF-confirmed or suspected)

ISSUE-3 · UT-00 SeedQuality — 2 cells

City	State	Strain	CSV	PDF	Action
Ashland	WI	Maple Presto	2.7	3.5	Patch to 3.5
Rosemount	MN	OT80-1	3	2.0	Patch to 2.0

ISSUE-4 · UT-00 SeedSize — 1 cell

City	State	Strain	CSV	PDF	Action
Fargo	ND	Maple Presto	14.2	13.3	Patch to 13.3

Note: SeedSize Fargo/Rosemount column swap suspected for some strains. Verify full SeedSize block before patching.

ISSUE-7 · PT-III — 2 cells

City	State	Strain	Trait	CSV	PDF	Action
Lafayette	IN	L77-443	Lodging	2.3	1.8	Patch to 1.8
Ottumwa	IA	HC76-3863	Oil	23.3	22.9	Patch to 22.9

ISSUE-9 · PT-IV Lexington Oil — 2 cells (*needs manual PDF verification*)

CSV matches XLSX exactly; discrepancy is between XLSX source and PDF printed report.

Strain	XLSX/CSV	PDF	Action
L77-8079	22.6	20.5	Open PDF ~p.194 PT-IV Lexington Oil column to confirm

Strain	XLSX/CSV	PDF	Action
L77-8209	22.1	21.4	Same

ISSUE-25 · Scattered cell errors — UT-0 through UT-IV (17 cells, needs XLSX verify)

Test	City	State	Strain	Trait	CSV	PDF
UT-0	Elora	ONT	M72-24	Protein	42.1	43.2
UT-I	Oakes	ND	Evans (0)	SeedQuality	1.0	2.5
UT-I	Ridgetown	ONT	Hardin	Protein	41.0	41.7
UT-I	Lamberton	MN	A78-121014	SeedSize	19.4	19.6
UT-I	Dekalb	IL	Evans (0)	YieldRank	10	9
UT-I	Dekalb	IL	M75-2	YieldRank	9	7
UT-II	Ridgetown	ONT	A78-227015	Lodging	2.8	1.2
UT-II	Ridgetown	ONT	A78-227016	Lodging	2.2	1.8
UT-II	Ames	IA	A78-122028	Protein	37.5	40.9
UT-II	Waseca	MN	A78-227013	Lodging	2.0	1.7
UT-II	Urbana	IL	Pella (III)	SeedQuality	3.3	2.0
UT-III	Elk Point	SD	U36276	SeedQuality	3.5	3.0
UT-III	Manhattan	KS	HW74-3385	Lodging	2.5	1.0
UT-IV	Portageville Clay	MO	K1033	SeedQuality	1.5	2.5
UT-IV	Queenstown	MD	Franklin	SeedQuality	2.2	1.2
UT-IV	Manhattan	KS	H76-3840	Protein	42.8	42.2
UT-IV	Manhattan	KS	H76-3840	Oil	21.5	19.1

Note: H76-3840 issues will be resolved once ISSUE-19 (HC76-3840 merge) is applied — verify post-merge.

CATEGORY D — Missing Data (extraction gaps)

ISSUE-8 · PT-I/III — NULL values linked to ghost strains

Most of these will resolve after ISSUE-5 (L77-709 → L78-709 OCR fix). Remaining genuine NULL:

Test	City	State	Strain	Trait	PDF value	Note
PT-I	Ridgetown	ONT	W442	Protein	41.5	Not linked to ghost strain
PT-III	Girard-Urbana	various	L78-709	Lodging, Height	various	Resolves after ISSUE-5

Test	City	State	Strain	Trait	PDF value	Note
PT-III	S. Charleston	OH	L77-709	Multiple	multiple	Resolves after ISSUE-5

ISSUE-10 · PT-IV / Portageville Loam_MO — Maturity missing for all 36 strains

Root cause: PT-IV Maturity block for southern locations is labeled **tp7** in XLSX (row 1620), not **tp8**. The extraction captured Eldorado/Queenstown/Lexington Maturity correctly but missed Portageville Loam entirely.

XLSX source data (tp7 block, row 1621):

- Reference: Union (IV) = 9/15 = DOY 259 (1980 leap year)
- All 36 strain offsets available in col 3 of that block

Action: Write `fixes/fix_pt4_portageville_maturity_1980.py` to compute and insert 36 Maturity DOY rows.

ISSUE-2 · UT-00 / Morden_MAN — SeedQuality NULL, all 9 strains (*no action*)

XLSX source has NULL for Morden SeedQuality for every strain. Morden did not report seed quality in 1980. Confirmed genuine missing data.

CATEGORY E — False Positives (no action needed)

FP-1 · Maturity off-by-1 DOY — widespread across all tests

Cause: Claude computes DOY using a non-leap calendar (Feb = 28 days). In 1980 (leap year), this produces a 1-day error for any date after Feb 28. CSV values computed from XLSX are correct. XLSX cross-checks confirm this pattern in UT-00, UT-0, UT-I, UT-III, UT-IV.

Pattern: Claude says `csv_value - 1` or `csv_value + 1`; XLSX confirms CSV is right. No fix needed.

FP-2 · PT-IV Maturity flags — Queenstown and Eldorado

Claude computed wrong DOY from PDF offsets (varied errors, not the uniform off-by-1 pattern). XLSX **tp7** block cross-check confirms CSV values are correct for all strains at both locations. No fix needed.

FP-3 · Ambiguous/matching noise

Claude flags these as discrepancies when values actually match:

- Integer vs decimal: **3** vs **3.0**
- Asterisk suffix: **9*** vs **9**
- Confirmed-matching values where Claude adds explanatory notes

Prompt improvement: Add to `VALUES_PROMPT_TEMPLATE`: "If the numeric value matches (ignoring trailing **.0** and ***** suffix), do NOT list it."

Pre-existing Open Items (from NUST_1980_Open_Items.md)

- **FLAG 1** — PlantingDate NULL for 5 locations — manual PDF lookup needed
- **FLAG 2** — Conductor NULL all 141 rows — not in 1980 publication
- **FLAG 3** — checksTable RM NULL all 26 rows — pre-RM era
- **FLAG 4** — 194 null-DOY rows — expected missing data

XLSX Source Structure Notes

UT-00 tp marker column order (confirmed)

tp marker	Table	Location column order
tp6	Yield (bu/a)	Fargo → Morden → Brandon
tp7	Yield Rank	Fargo → Morden → Brandon
tp8	Maturity	Fargo → Morden → Brandon
tp9	Lodging	Fargo → Morden → Brandon
tp10	Height	Morden → Brandon → Fargo ← swapped
tp11a	SeedQuality	Morden → Brandon → Fargo ← swapped
tp11b	SeedSize	Morden → Brandon → Fargo ← swapped
tp12a	Protein	Ottawa, Elora, Ashland, Morden, Rosemount only
tp12b	Oil	Ottawa, Elora, Ashland, Morden, Rosemount only

PT-IV tp7 Maturity block (row 1621, file 2)

Covers: Portageville Loam, Portageville Clay, Belleville, Eldorado, Queenstown, Lexington (6 southern locations). Labeled **tp7** instead of **tp8** — extraction captured all locations except Portageville Loam (see ISSUE-10).